SUBSEA MATTRESS VERIFICATION REPORT: CR-A1_M15 (#A046)

Project: Tung-Hsiao Power Plant 2nd Stage Renewal Subsea Gas Pipeline Project
Client: Taiwan Power Company (TPC) | Contractor: NMDC Energy | Vessel: DP2 MMA Prestige
Crossing Array: CR-A1_L1 | Location ID: CR-A1_M15 | Mattress ID: #A046 | Active USBL: Survey Pole
Survey Date/Time: 2026-11-04 14:34 | Deliverable Base: 20261104_Mattress_CR-A1_M15_A046
Chart Scales @ A3: Panel 1 (1:250) | Panel 2 (1:75) | Panel 3 (1:75/1:15) | Panel 4 (1:50/1:15)

EXECUTIVE VERDICT

- Overall Status: VERIFIED - WITHIN DESIGN & GAP TOLERANCE
- Identification & Sensor: Mattress #A046 at CR-A1_M15 (CR-A1_L1) tracked via Survey Pole.
- Centerline Offset (dCL): 0.265 m vs. Tolerance +/-0.50 m (PASS) [dAlong: +0.050 m, dCross: -0.260 m].
- Heading Alignment: Design South Brg 195.854 deg vs. As-Laid 196.889 deg (dBrg: +1.035 deg).
- ROV Corner Gaps: 1st Unit (N/A) | Max Gap 0.00 m vs. Tolerance <=0.20 m (N/A (1st Unit)).

1. GEODETIC, 4-CORNER ROV & GAP COMPARISON TABLE

Parameter	Design Baseline	QA/QC USBL Fix	As-Laid 4-Corner ROV	Delta / Status
Easting E (m, TWD97)	194541.443	194541.206	194541.206	-0.237 m
Northing N (m, TWD97)	2702613.394	2702613.513	2702613.513	+0.119 m
Depth Z (m, TWVD2001)	Seabed + 0.30 m	-	-18.20	Verified WROV
South Bearing (deg)	195.854 deg	196.889 deg	196.889 deg	+1.035 deg
North Azimuth (deg)	15.854 deg	-	16.889 deg	+1.035 deg
Radial Offset dCL (m)	0.000 m (Tol +/-0.50m)	-	0.265 m	PASS (87.4% in box)
ROV Corner Gaps (m)	Tol <=0.20 m	-	1st Unit (N/A)	N/A (1st Unit)

2. KEY SURVEY FINDINGS (4-CORNER ROV COORDINATES & ENVIRONMENT)

- C1 (NW): E 194540.642 m, N 2702616.819 m, Z -18.20 m | ROV Gap NW: N/A
- C2 (NE): E 194543.513 m, N 2702615.948 m, Z -18.20 m | ROV Gap NE: N/A
- C3 (SE): E 194541.770 m, N 2702610.207 m, Z -18.20 m | ROV Gap SE: N/A
- C4 (SW): E 194538.899 m, N 2702611.078 m, Z -18.20 m | ROV Gap SW: N/A
- Edge Clearances: NNE +0.08m (Nbr: CR-A1_M8), SSW -0.02m (Nbr: CR-A1_M22), ESE -0.21m (Nbr: CR-A1_M16), WNW + 0.32m (Nbr: CR-A1_M14).
- Environment: Wind: Recorded in Log | Current: Recorded in Log.

3. OPERATIONAL CRANE / ROV NUDGE RECOMMENDATION

- Required Vector to Design Center: Shift 0.27 m toward ESE (Azimuth 116.7 deg).
- Cartesian Adjustments: dE = +0.237 m, dN = -0.119 m.

4. SURVEYOR REMARKS

- [Survey Pole] Mattress installed completed (CR-A1_M15) #A046 (14:34 - 16:00:20); dCL=0.265m, dBrg=-1.028 deg

5. HISTORICAL CROSSING LEDGER

Unit #	Date/Time	Location ID	Tag ID	As-Laid E	As-Laid N	Brg (deg)	dCL (m)	Gaps
1	2026-11-04 14:34	CR-A1_M15	#A046	194541.206	2702613.513	196.89	0.265	1st Unit (N/A)
2	2026-11-05 04:28	CR-A1_M16	#A007	194544.356	2702612.753	198.40	0.180	NW:0.02m SW:0.10m